

Supply Chain Database Using SQL

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Table of Contents

Introduction.....	3
Project Summary.....	4
Project Objectives	5
Project Overview	5
Database Design.....	6
Suppliers.....	6
Projects	6
Purchase Requisitions	7
Purchase Orders.....	7
Post Order Expediting (POSE).....	8
Freight Companies	8
Freight Estimations	9
Figure 1	10
Entity Relationship Diagram (ERD) of the Supply Chain Procurement Database	10
SQL Implementation.....	11
Figure 2	11
Sample of the SQL implementation (<i>Database.sql</i>) showing table creation and sample data insertion.	11
Figure 3	12
Sample of the SQL queries (<i>SQLQueries.sql</i>) used to retrieve and analyze data from the database.	12
Project Files	13
Conclusion	14

Introduction

During my internship in the **Supply Chain Department** at **Saipem**, I was introduced to several procurement and logistics processes that support large engineering projects. While I gained practical experience using systems such as **SAP** and worked with procurement documents, supplier information, logistics activities, and post-order expediting, I wanted to strengthen my technical skills by connecting this experience with **SQL**.

This project was created to simulate a simplified procurement database inspired by real business processes I observed during my internship. The database demonstrates how **SQL** can be used to organize, retrieve, and analyze procurement data through a relational database structure. It combines my understanding of supply chain operations with my interest in databases and SQL, allowing me to apply technical skills to a realistic business scenario.

Project Summary

This project presents the design and implementation of a relational Supply Chain Database using SQL. It was developed as a personal portfolio project inspired by my internship experience in the Supply Chain Department at Saipem. During my internship, I gained exposure to procurement, supplier management, purchase requisitions, purchase orders, logistics, freight estimations, post-order expediting (POSE), and SAP-based business processes. While my internship primarily focused on understanding these operational workflows, this project provided an opportunity to connect that experience with my interest in SQL and database management.

The objective of this project was to design a structured database that models key supply chain activities using realistic business scenarios and sample data. After creating and populating the database with sample data, a series of SQL queries were developed to retrieve, filter, summarize, and analyze procurement information. These queries demonstrate fundamental SQL concepts, including data retrieval, filtering, joins, aggregate functions, and grouping, while producing meaningful business insights from the database.

This project demonstrates my ability to translate real-world business processes into a relational database design and apply SQL to solve practical business problems. More importantly, it reflects how I combined knowledge gained during my internship with my technical skills to build a portfolio project that showcases my growing interest in database management and data analytics.

Project Objectives

- Design a relational database using SQL.
- Apply database concepts such as primary keys and foreign keys.
- Simulate procurement and logistics activities using realistic business data.
- Practice writing SQL queries to retrieve meaningful business information.
- Demonstrate how SQL can support decision-making within supply chain operations.

Project Overview

This project models a simplified procurement process inspired by my internship experience in the Supply Chain Department at Saipem. The database represents procurement activities for two Saudi Aramco projects:

- ***Jafurah Gas Development Project***
- ***Berri Offshore Fields***

The database includes supplier information, purchase requisitions, purchase orders, freight companies, freight estimations, and Post Order Expediting (POSE) activities. Although the data used in this project is fictional, it was designed to reflect realistic procurement workflows while demonstrating SQL concepts in a practical business context.

Database Design

The database consists of *seven* relational tables designed to simulate procurement and logistics activities inspired by my internship experience. Each table stores a specific type of information and is connected through primary and foreign keys to maintain data integrity.

Suppliers

Purpose: The **Suppliers** table stores information about companies that provide products or services for procurement activities.

Column	Data Type	Description
SupplierID	INTEGER	Unique identifier for each supplier (Primary Key)
SupplierName	TEXT	Name of the supplier
Country	TEXT	Country where the supplier is located
Category	TEXT	Type of products or services provided

Projects

Purpose: The **Projects** table stores information about the projects associated with procurement activities.

Column	Data Type	Description
ProjectID	INTEGER	Unique identifier for each project (Primary Key)
ProjectName	TEXT	Name of the project
Client	TEXT	Client associated with the project
Location	TEXT	Project location

Purchase Requisitions

Purpose: The **PurchaseRequisitions** table records material requests submitted by different departments before purchase orders are created.

Column	Data Type	Description
PR_ID	INTEGER	Unique purchase requisition number (Primary Key)
ProjectID	INTEGER	References the related project (Foreign Key)
RequestDate	DATE	Date the requisition was created
Department	TEXT	Department requesting the material
MaterialDescription	TEXT	Description of the requested material
RequestedBy	TEXT	Employee or role requesting the material
Status	TEXT	Current status of the requisition

Purchase Orders

Purpose: The **PurchaseOrders** table stores purchase orders issued to suppliers after purchase requisitions are approved.

Column	Data Type	Description
PO_ID	INTEGER	Unique purchase order number (Primary Key)
PR_ID	INTEGER	References the related purchase requisition (Foreign Key)
SupplierID	INTEGER	References the supplier (Foreign Key)
OrderDate	DATE	Date the purchase order was issued
Amount	DECIMAL(10,2)	Total purchase order amount
Currency	TEXT	Currency used for the purchase order
Status	TEXT	Current purchase order status
SABERStatus	TEXT	Status of the SABER certification requirement

Post Order Expediting (POSE)

Purpose: The **PostOrderExpediting** table tracks supplier follow-up activities after purchase orders have been issued.

Column	Data Type	Description
ExpeditingID	INTEGER	Unique identifier for each expediting record (Primary Key)
PO_ID	INTEGER	References the related purchase order (Foreign Key)
FollowUpDate	DATE	Date of supplier follow-up
ExpectedDelivery	DATE	Planned delivery date
ActualDelivery	DATE	Actual delivery date (if completed)
DeliveryStatus	TEXT	Current delivery status
Expeditor	TEXT	Person or team responsible for follow-up
Remarks	TEXT	Additional comments or notes

Freight Companies

Purpose: The **FreightCompanies** table stores information about freight service providers responsible for transporting materials.

Column	Data Type	Description
FreightCompanyID	INTEGER	Unique identifier for each freight company (Primary Key)
CompanyName	TEXT	Name of the freight company

Freight Estimations

Purpose: The **FreightEstimations** table stores estimated transportation information for purchase orders.

Column	Data Type	Description
EstimationID	INTEGER	Unique identifier for each freight estimation (Primary Key)
ProjectID	INTEGER	References the related project (Foreign Key)
PO_ID	INTEGER	References the related purchase order (Foreign Key)
FreightCompanyID	INTEGER	References the freight company (Foreign Key)
TransportMode	TEXT	Transportation method (Sea, Land, etc.)
Origin	TEXT	Shipment origin
Destination	TEXT	Shipment destination
EstimatedCost	DECIMAL(10,2)	Estimated transportation cost
EstimatedTransitDays	INTEGER	Estimated delivery time in days

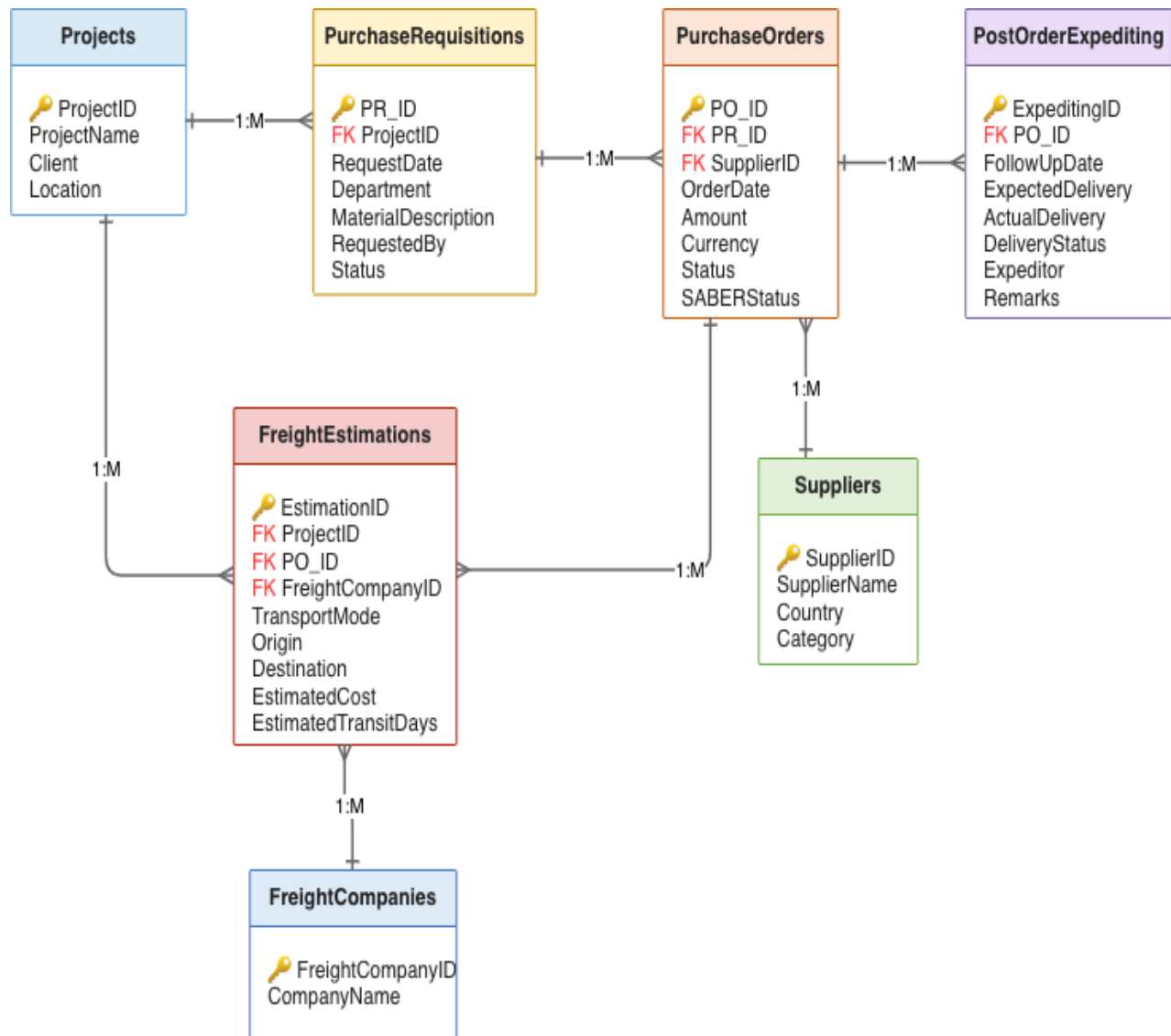


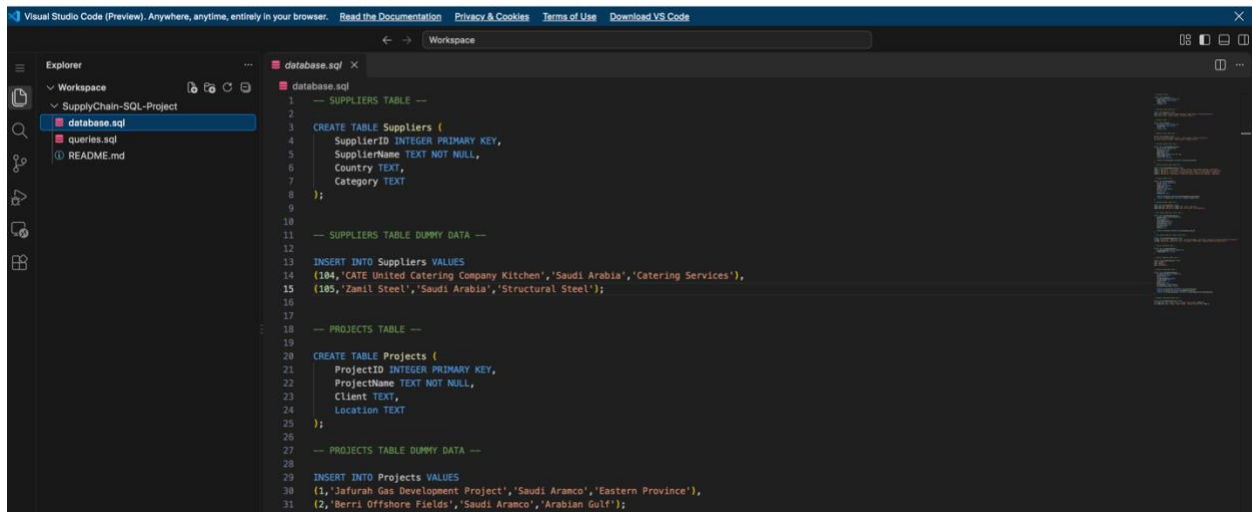
Figure 1.

Entity Relationship Diagram (ERD) of the Supply Chain Procurement Database

SQL Implementation

This project was developed using **SQLite** Online to design and implement a relational supply chain database. **SQL Data Definition Language (DDL)** statements were used to create the database schema, while **SQL Data Manipulation Language (DML)** statements were used to populate the tables with realistic sample data.

The database consists of seven related tables connected through primary and foreign keys to maintain data integrity and accurately represent supply chain workflows. **Figures 2 and 3** present samples of the SQL scripts used to create the database structure and insert the sample data. The complete implementation is provided in the supporting file **Database.sql**.



```
1  -- SUPPLIERS TABLE --
2
3  CREATE TABLE Suppliers (
4      SupplierID INTEGER PRIMARY KEY,
5      SupplierName TEXT NOT NULL,
6      Country TEXT,
7      Category TEXT
8  );
9
10
11 -- SUPPLIERS TABLE DUMMY DATA --
12
13 INSERT INTO Suppliers VALUES
14 (104, 'CATE United Catering Company Kitchen', 'Saudi Arabia', 'Catering Services'),
15 (105, 'Zamil Steel', 'Saudi Arabia', 'Structural Steel');
16
17
18 -- PROJECTS TABLE --
19
20 CREATE TABLE Projects (
21     ProjectID INTEGER PRIMARY KEY,
22     ProjectName TEXT NOT NULL,
23     Client TEXT,
24     Location TEXT
25 );
26
27 -- PROJECTS TABLE DUMMY DATA --
28
29 INSERT INTO Projects VALUES
30 (1, 'Jafurah Gas Development Project', 'Saudi Aramco', 'Eastern Province'),
31 (2, 'Berri Offshore Fields', 'Saudi Aramco', 'Arabian Gulf');
```

Figure 2.

Sample of the SQL implementation (*Database.sql*) showing table creation and sample data insertion.

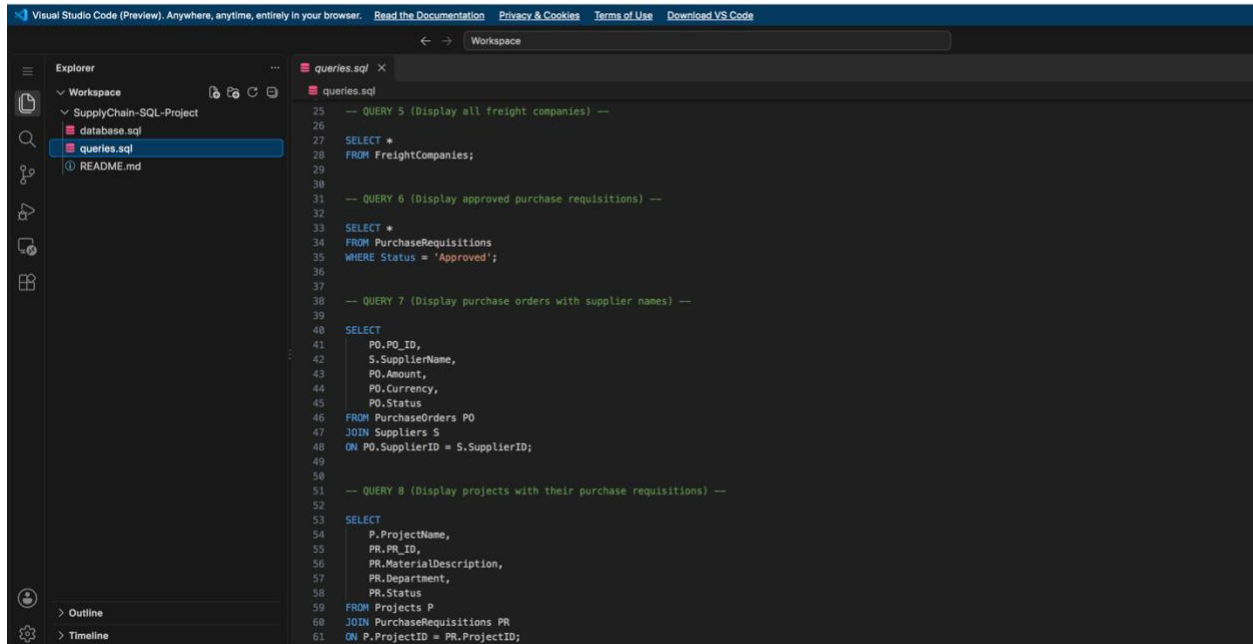
A screenshot of the Visual Studio Code editor interface. The top bar shows the Visual Studio Code logo and the text "Visual Studio Code (Preview). Anywhere, anytime, entirely in your browser." followed by links for "Read the Documentation", "Privacy & Cookies", "Terms of Use", and "Download VS Code". The Explorer sidebar on the left shows a project named "SupplyChain-SQL-Project" with files "database.sql", "queries.sql", and "README.md". The "queries.sql" file is selected and open in the main editor. The editor displays SQL queries with line numbers from 25 to 61. The queries are:
25 -- QUERY 5 (Display all freight companies) --
26
27 SELECT *
28 FROM FreightCompanies;
29
30
31 -- QUERY 6 (Display approved purchase requisitions) --
32
33 SELECT *
34 FROM PurchaseRequisitions
35 WHERE Status = 'Approved';
36
37
38 -- QUERY 7 (Display purchase orders with supplier names) --
39
40 SELECT
41 PO.PO_ID,
42 S.SupplierName,
43 PO.Amount,
44 PO.Currency,
45 PO.Status
46 FROM PurchaseOrders PO
47 JOIN Suppliers S
48 ON PO.SupplierID = S.SupplierID;
49
50
51 -- QUERY 8 (Display projects with their purchase requisitions) --
52
53 SELECT
54 P.ProjectName,
55 PR.PR_ID,
56 PR.MaterialDescription,
57 PR.Department,
58 PR.Status
59 FROM Projects P
60 JOIN PurchaseRequisitions PR
61 ON P.ProjectID = PR.ProjectID;

Figure 3.

Sample of the SQL queries (*SQLQueries.sql*) used to retrieve and analyze data from the database.

The complete SQL implementation and all SQL queries are provided in the supporting files *Database.sql* and *SQLQueries.sql*.

Project Files

File	Description
Database.sql	SQL script for creating tables and inserting sample data
SQLQueries.sql	SQL queries used to retrieve and analyze data
SQLQueriesOutputs.pdf	Screenshots of SQL query results
ER_Diagram.png	Entity Relationship Diagram
Project Report.pdf	Project documentation

Conclusion

This project allowed me to apply SQL to a realistic business scenario inspired by my internship experience in the Supply Chain Department at Saipem. By designing a relational database, creating sample datasets, and writing SQL queries, I strengthened my understanding of database design, table relationships, and business data analysis.

The project demonstrated how SQL can be used to organize and analyze supply chain data related to suppliers, purchase requisitions, purchase orders, freight estimations, and post-order expediting activities. Developing this database improved my confidence in applying SQL to practical business problems while reinforcing the importance of structured data in supporting operational decision-making.

Overall, this project provided an opportunity to connect my internship experience with my interest in database management and data analytics, creating a portfolio project that reflects both my technical skills and my understanding of real-world supply chain processes.